8. PD16 — Alex meeting takeaways (Aug 4, 2026)

Last synced: 2026-08-04
Audience: Charles, re-entering after the Phase B characterization briefing.
Standalone: definitions inline; no scavenger hunt required.
Source transcript: `transcripts/20260804_Paper_Directions_16_otter_ai_transcript.docx`
Working digest (agents): `transcripts/PD16_notes.md`
Deck briefed: `HEROs_and_PASSes/slides/CHAR_Phase_B_characterization.pptx` (+ walkthrough doc 07).

Headline — what changed in one paragraph

Alex liked the **characterization deck** (intro, ρ, λ, θ, θ×K/N, γ, λ_crit). The meeting did **not** ask you to throw away Phase B. It **re-aimed two definitions** and opened a **calibration path**:

1. **L_C** should be a **team** congestion measure (how many viable peers on the roster), not a leave-one-out per-player variant — at least for the score story Alex wants next.
2. **θ** should come from **K/N and the ability draw** (naive draft / no teams): the ability cutoff where the top **K/N** tail of the talent distribution lives — not a free 539 preset like 0.72.
3. **ρ** gets a sharper interpretation: it **dials how spread out team L_C values are** across the league (your whiteboard histograms), not only "the selection curve bends."

Only **γ** and **λ** remain as knobs to fit from data once ρ and θ are pinned — and Alex wants that fit to use **properties of the data**, not "slide the sim curve until it matches the hero."

What Alex accepted (keep saying this)

Topic	Status
Phase B = **characterization**, not NCAA curve fitting	✓
Pipeline: ASSIGN → SCORE (**S_i = A_i − λ L_C**) → SELECT (top-K) → VISUALIZE	✓
Smooth L_C in deck; hard share still in code	✓
λ in score bends selection vs talent-only	✓
θ × K/N grid worth keeping; revisit after θ redefinition	✓
Sort-and-chop for γ / **λ_crit ≈ 4/γ** — deterministic testbed	✓ (not rejected)

Whiteboard A — ρ dials the distribution of team L_C

You drew three panels; Alex confirmed the story.

1D: # teams vs L_C

ρ	Shape of histogram
Low (little assortativity)	One **narrow** hump — almost a **spike** (Alex: not even uniform; teams look similar)
High (strong assortativity)	**Spread** — weak teams pile up near **L_C ≈ 0**, strong teams at **high L_C**

Plain English: ρ controls **how much team congestion varies** across the league. Low ρ → everyone faces similar peer pressure; high ρ → elite rosters are crowded, weak rosters are not.

2D heatmap: team ability vs L_C

- **x:** team **L_C**
- **y:** team ability (mean **A** on roster, or target **T_j**)
- **Color:** number of teams in each cell

Question answered: Do better teams have more congestion? **Yes** — expect an upward cloud (your wiggly diagonal in the oval).
Why this matters: Gives ρ a **model-internal** interpretation Alex can cite, beyond "the selection-by-pool-mean plot moved."
Not built yet: scripts/figures for this diagnostic (checklist item in PD16_notes).

Whiteboard B — naive draft defines θ

Setup: Draw **A_i** from an ability distribution (Beta(2,2) on [0,1] in the sim; empirical in real data).
Thought experiment: No teams — everyone competes on talent alone. Exactly fraction **K/N** of the league advances (naive draft).
Definition:
Find ability cutoff **A*** such that

$$P(A_i > A^*) = K/N$$

(i.e. the **right-tail area** under the ability PDF equals **K/N**).

Set θ = A*.

Meaning: θ is the talent threshold where, **if roster ambiguity did not exist**, everyone above would advance and everyone below would not. The **sigmoid** $\sigma(\gamma(A_j - \theta))$ then adds **graded** peer viability around that threshold — ambiguity at the margin, not a hard cliff.

Today vs PD16:

	Today (Phase B deck)	PD16 direction
θ at K/N = 10%	Fixed **0.72** (539 preset)	**Computed** from Beta + K/N (≈ **0.68** for Beta(2,2) at 10%)
θ when K/N changes	Grid varies K/N but θ arms were {0.50, 0.72, 0.90}	θ **tracks** K/N by construction when using naive-draft mode

Not implemented yet in gallery scripts — planned as an opt-in switch (see below).

L_C: team-level vs LOO (Alex ask)

	LOO L_C (today)	Team L_C (PD16)
Unit	One value per **player** *i*	One value per **team** *j*
Peers	Teammates **excluding** *i*	Whole roster (all teammates)
Story	"My congestion field"	"How crowded is this team?"
Alex	Fine for hero axis; maybe over-customized in **score**	Matches "congestion = good players on the team"

Alex noted early runs that "worked" used **poolq**-style team context, not LOO, for the congestion-in-score story. He thinks team-level L_C will **come out in the wash** for rankings but wants you to **try it and compare**.

Empirical calibration roadmap (end of meeting)

Knob	From data?	How (PD16)
A_i distribution	Yes	Empirical perf or declared draw
K/N	Yes	Domain (MBB ~1%, gallery 10%, etc.)
θ	Yes	Quantile: $\boldsymbol{\theta = F_A^{-1}(1 - K/N)}$
ρ	Yes	Match assignment assortativity in panel
γ, λ	**Open**	Need principled estimators — **not** "match the hero curve" as the definition

"Behind an environment flag" — what that literally means

Not a new **Anaconda / conda environment.** Same Python, same `conda activate` you use now.

Yes — a **shell environment variable** (toggle) read by `sports/scripts/gallery_knobs.py`, the same way the repo already does:

```
export GALLERY_PRESET=539
export GALLERY_HERO_SEED=42
export GALLERY_K_OVER_N=0.10
./scripts/build_characterization_slides.sh
```

Planned PD16 toggles (names tentative until implemented):

Variable	Default (unchanged deck)	PD16 experimental
`GALLERY_LC_MODE`	`loo_smooth`	`team_smooth`
`GALLERY_THETA_MODE`	`preset` (0.72 from 539)	`k_over_n` (quantile from ability draw + K/N)

Why flags: Running `./scripts/build_characterization_slides.sh` with **no** exports keeps producing **today's PNGs** for your hand deck. You opt in when ready:

```
export GALLERY_LC_MODE=team_smooth
export GALLERY_THETA_MODE=k_over_n
./scripts/build_characterization_slides.sh
```

Then **compare** peak bins, curve shapes, and θ×K/N cells to the current deck — evidence before you change slide footers or benchmarks.

Open work (priority)

1. **Code:** team L_C + θ(K/N) toggles in `gallery_knobs.py` + pool assignment path; compare figures.

2. **Figures:** L_C distribution vs ρ (whiteboard A) — histogram + 2D heatmap.
3. **ρ slide:** try $\rho \to 0.001$; one clean sentence on why ρ matters in this model.
4. **Intro / doc 07:** update θ provenance when naive-draft θ becomes default.
5. **Later:** principled **γ, λ** from data; Menger multiplicative congestion (parked).

Sort-and-chop (unchanged role)

Still the **γ / λ_crit** testbed: extreme assignment removes overlap so **λ_crit ≈ 4/γ** is readable. PD16 congestion changes are **orthogonal** — Alex wants both.

Quotes worth keeping

- **ρ:** "Assortativity is this dialing measure of what's the spread of the L_Cs."
- **θ:** "In a world where there was no team congestion, the cutoff … pick θ so that area = K/N."
- **L_C:** "Congestion should be how many good players are on the team."
- **Fit:** "Try to find γ and λ without … just make the two curves as close as possible."

Related files

Doc	Role
`07_Phase_B_Characterization_Slides_Explained.md`	Deck walkthrough (pre-PD16 θ/L_C)
`06_Lambda_threshold_and_KN_memo.md`	λ_crit, sort-and-chop
`CHARLES_CHECKLIST.md`	Near-term tasks
`transcripts/PD16_notes.md`	Agent-oriented digest + checklist